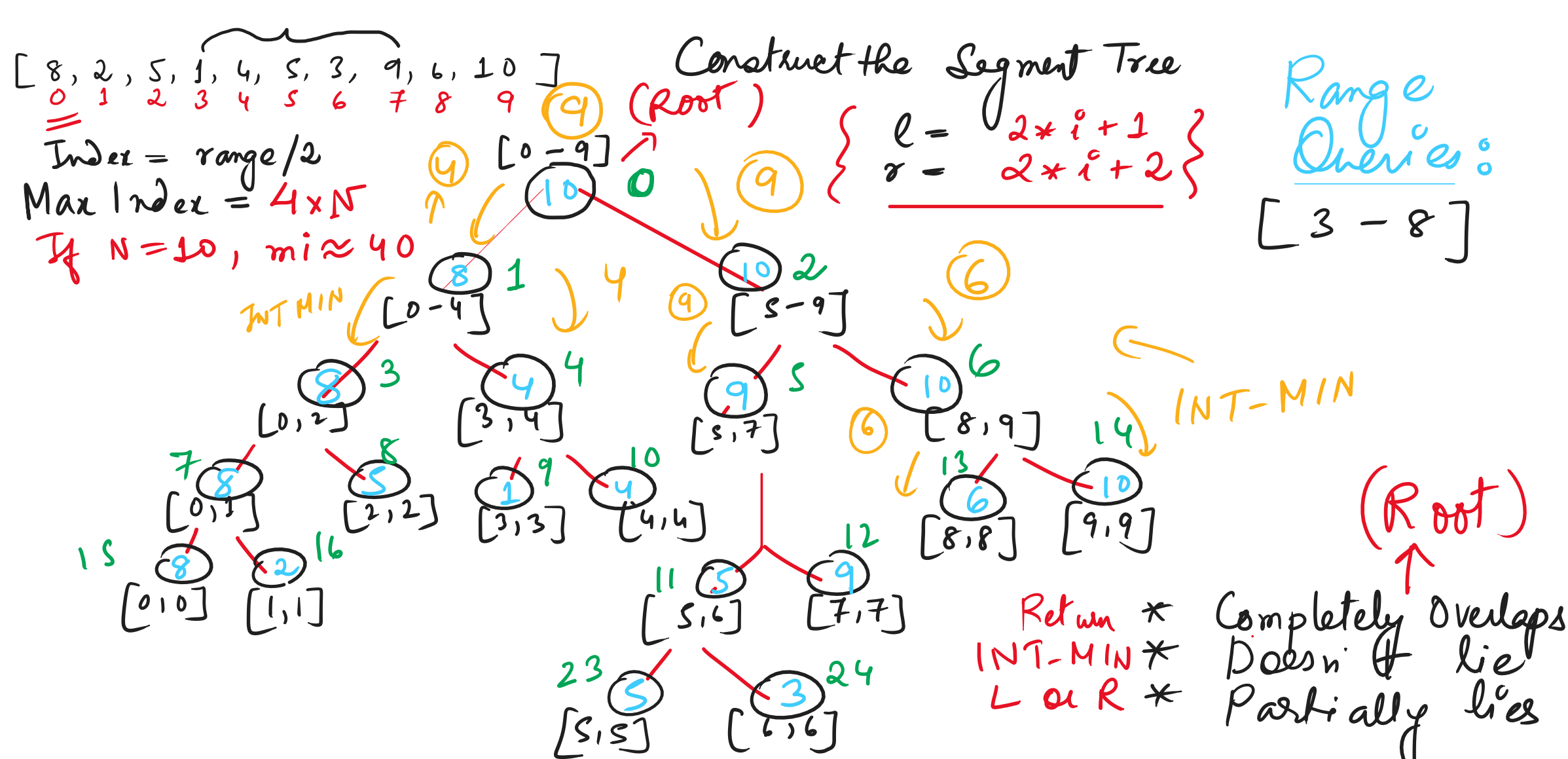


(Segment Trees) [Range based Search Queries]
 [Important for certain (array based) Interviews]

Suppose you are given an array: $\rightarrow$
 $[8, 2, 5, 1, 4, 5, 3, 9, 6, 10]$ $O(N \times Q)$ (Slow)

And you have to find the max number in a range:

* Number of Queries: $\rightarrow Q = 3$
 $mx [0, 3]$ $mx [4, 6]$ $mx [3, 8]$
 $Q1$ $Q2$ $Q3$



LRU Cache [Interviewer's Favourite Topic in OS]
 Least Recently Used Cache

Best Example: [Front Door Shoe Rack]

* Imagine you have a small shoe rack right by your front door. Suppose it can only hold exactly 3 pairs of shoes (this is your Cache Capacity)

* The rest of your shoes are upstairs in a big closet (almirah) [This is your Main Memory or Database]

* Walking upstairs takes time so you want to keep the shoes you wear most often right at the front door.

* Here is how a person's life works with this shoe

1. Mon $\rightarrow$ You wear Sneakers. You put back on rack. Rack: Sneakers

2. Tue $\rightarrow$ You wear Boots. You put back on rack. Rack: Sneakers, Boots

3. Wed $\rightarrow$ You wear Sandals. You put back on rack. Rack: Sneakers, Boots, Sandals

4. Thu $\rightarrow$ You buy a brand new Loafers and want to put on the rack.

As you the rack is full! You have to take one pair upstairs to the big almirah to make room for the Loafers.

Question? Which pair do you remove?

If you are using an LRU Cache, you remove the shoes that you haven't used/touched in the longest amount of time.

* You just wore Sandals on Wed

* You wore Boots on Tue

* You haven't worn the Sneakers since Monday.

So, the Sneakers are the "Least Recently Used" You kick them off & put Loafers.

Cache: A small, super-fast memory storage where all computers keep data they think will be needed again soon, so that they don't have to fetch it from the slow main database.

LRU: (Least Recently Used) rule ("eviction policy") The specific the computer uses to decide what to delete when the fast memory gets full. It deletes the data which hasn't been used/accessed in the longest time.

Java: Linked Hash Map $\rightarrow$ capacity